Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/26/2005
PI Name:	Goree, John Ph.D.		
Project Title:	Optically-Excited Waves in 3D Dusty Plasmas		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	04/01/2002	**End Date:**	03/31/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2771		
Performance Goal No.:			
Performance Goal Text:			
	Original Abstract: Experiments will be carried out in an interdisciplinary study of waves (i.e., phonons) in 3D Coulomb lattices. The experimental system is a suspension of charged polymer microspheres in a gas-discharge plasma. This so-called "dusty plasma" is a macroscopic system where the microspheres arrange themselves, due to their mutual repulsion, in a lattice. Shear waves will be excited in the lattice using the radiation pressure of a laser beam. New Strategic Abstract: During the final two years of this grant, the research goals are shifted to emphasize strategic goals related to NASA's planned manned space flight missions to the Moon and Mars. Countermeasures for dust contamination in Advanced EVA will be the theme. Experiments will be performed to test materials and investigate processes related to dust adhesion on surfaces and the use of plasma exposure to remove dust from those surfaces. Original Objective: We will prepare for future flight experiments, which will have two tasks: * We will study the		

dispersion relation (i.e., phonon spectrum) of sinusoidal waves and the propagation of pulsed waves. These experiments will be performed with plane waves. * We will study the formation of Mach cones, which are the V-shaped wakes or shocks created by a supersonic disturbance. In this experiment, a moving laser spot is the supersonic disturbance.

New Strategic Objective:

Preliminary experiments will be carried out to investigate adhesion properties of JSC-1 Lunar Simulant dust to EVA suits, and the effectiveness of plasmas in removing that dust.

Original description: * Need for microgravity: These experiments are an extension of 2D experiments performed in the laboratory, under gravity conditions. Sedimentation due to gravity causes the microspheres to settle rapidly into a 2D layer that is levitated by the strong electric field near an electrode in a plasma. By eliminating gravity, it is possible to fill a 3D volume with the suspension. * Use of a novel experimental system: Our experiments will exploit the low damping of a dusty plasma as compared to colloidal suspensions. We will excite waves using a manipulation laser, which applies a radiation pressure on the particles. Using a dusty plasma allows us to measure particle motion in-situ using direct imaging with video microscopy. * Our scientific effort and our team are interdisciplinary.

New Strategic Task Significance:

Advanced EVA has been identified as one of the four critical technologies for NASA's New Vision for Space Exploration. The EVA System includes EVA suits and airlocks, for which dust contamination is a risk. According to Lara Kearney, JSC's manager of Advanced EVA, 13 challenges of present-day technologies have been identified, and two of these include dust: the present-day EVA suit protection from dust intrusion is inadequate, and dust on airlocks is an unsolved risk.

Our project will include studies for low-TRL technology needs related to dust on the Moon and Mars.

Task Description:

"Iowa Dust Mitigation Scheme": We are performing experiments at The University of Iowa to test methods of cleaning space suits. In our "Iowa Dust Mitigation Scheme," the crew of a lunar habitat would use two steps to clean their suits after returning to the habitat from an EVA. Firstly, they will use air to blow dust off the suits during airlock pressurization, and secondly, after egress from the suit, they will use plasma to electrostatically clean the residual dust.

These two steps have the following advantages: (1) they can both be done in the airlock, with dust collecting at the bottom of the airlock, thereby isolating it from entering the habitat; (2) they will not require any water, which will be a precious resource; (3) they do not involve any frictional rubbing of the suit's fabric, which would reduce the suit's lifetime.

1st step of the "Iowa Dust Mitigation Scheme": We will exploit the availability of air flow during pressurization of the airlock. We propose that air should not be introduced into the airlock diffusely, but instead it should be introduced through a hose that the astronaut will direct onto the spacesuit, while still wearing the suit. This will blow off a large fraction of the dust, which would collect at the bottom of the airlock where it can be emptied periodically.

2nd step of the "Iowa Dust Mitigation Scheme": The second step will exploit the availability of vacuum conditions in an airlock. After the astronaut has removed the space suit and egressed, the suit will hang in the airlock. The airlock will then be partially evacuated, with a gas pressure of approximately 1 mBar. Radio-frequency electrical power will be applied, ionizing the gas to make plasma that will be exposed to the airlock's contents, including the suit. Electrostatic effects will remove dust from the suit. This method was first introduced by T.E. Sheridan et al., J. Geophys. Res. 97, 2935 (1992), and it has been used in the semiconductor industry for cleaning particulates from silicon wafers.

While our scheme is intended to be used in a lunar Habitat, it could also be adapted to a habitat on Mars, which will also require airlocks.

The electrostatic effects that will remove dust from the suit will be the primary focus of this research. When a particle rests on a surface, it is held in place by a van-der-Waals attraction, but this adhesion can be overcome by several mechanisms. One such mechanism for the Moon and Mars environments is the application of electrostatic forces.

Scientific results from this project will also find other applications: (1) One issue is cleaning solar panels on Mars. Dust storms on Mars will gradually cover the panels, reducing their efficiency, unless they are cleaned. Because Mars has a very low-pressure atmosphere, it may be possible to use radio-frequency power to ionize the Martian atmosphere immediately above a solar panel, and thereby clean it. (2) Another issue is gaining an understanding of dust particle leviation and transport on the Moon. The Moon, unlike Mars, has no atmosphere. Instead, it is immersed in interplanetary plasma, which consists of free electrons and protons at a low number density. The Moon disturbs this plasma, so that there is an electric field perpendicular to the Moon's surface. This electric field applies a vertical force to any electrically-charged fine particle. Particles are known to become charged and levitated on the Moon. This was discovered in Apollo missions, where a thin cloud of dust was seen floating above the lunar horizon. Particles can be charged either photoelectrically by exposure to solar UV, or by collection of free electrons and ions from the surrounding plasma. This can lead to levitation, if the particle can first overcome van-der-Waals adhesion to surfaces such as a rock, a photovoltaic cell, or another dust particle.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

A number of laboratory experiments were reported in the following publications, along with supporting numerical simulations and theory:
Refereed Journals:

K. Avinash, P. Zhu, V. Nosenko and J. Goree "Nonlinear Compressional Waves in a Two-Dimensional Yukawa Lattice" Physical Review E, Vol. 68, 046402-1 - 046402-8 2003

V. Nosenko, J. Goree, Z.W. Ma, D.H.E. Dubin and A. Piel "Compressional and Shear Wakes in a 2D Dusty Plasma Crystal" Physical Review E Vol. 68, 056409-1 – 056409-15, 2003

B. Liu, K. Avinash and J. Goree "Transverse Optical Mode in a One-Dimensional Chain" Physical Review Letters, Vol.

	91, 255003, 2003 V. Nosenko, K. Avinash, J. Goree and B. Liu "Nonlinear Interaction of Compressional Waves in a 2D Dusty Plasma Crystal" Physical Review Letters, Vol. 92, 085001-1– 085001-4, 2004 B. Liu, K. Avinash and J. Goree "Characterization of Interparticle Potential from Lattice Structure" Physical Review E, Vol. 69, 036410, 2004 Robert L. Merlino and John A. Goree Dusty Plasmas in Industry, the Laboratory and Space Physics Today, Vol. 57, 32-38, July 2004 V. Nosenko and J. Goree Shear Flows and Shear Viscosity in a Two-Dimensional Yukawa System (Dusty Plasma) Physical Review Letters, Vol. 93, pp. 155004-1 - 155004-4, 2004
Task Progress:	In addition to the papers published, the following work was performed: Parabolic flight experiment: Hardware construction was completed for a parabolic flight rig. The rig was flown twice in October 2003 aboard NASA's KC-135. The immediate objective of the experiment is to observe voids in a 3D suspension in a dusty plasma, imaging both the particle suspension and the glow. It was found that the glow is enhanced when particles are present as compared to when the plasma has no particles, and the glow is enhanced most strongly in the void. Further data analysis is now underway. Longer-term objectives for this experiment were to develop hardware that will be used in the future for testing experimental concepts and components for space flight. Preparations for flight experiments: On November 28, 2003, the PI participated in an ESA meeting at ESTEC, in the Netherlands, via videoconferencing. The purpose of the meeting was to capture user requirements for the IMPF facility of ESA's IMPACT laboratory, which is now being designed by an ESA contractor. The PI provided design requirements for the future flight experiments that are intended to develop from the current project. These experiments include three kinds of waves: naturally-occurring phonons, optically-excited sinusoidal sound waves, and optically-excited Mach cones. Before and after the meeting, the PI has had significant discussions with the systems engineer for ESA's contractor, Kayser Threde, for this design. The PI's involvement in this design was requested by ESA. The contact person at ESA is Dr. Frank Molster, Frank.Molster@esa.int Dust mitigation experiments: We performed a preliminary experiment demonstrating that a plasma can remove JSC-1 Lunar Simulant dust from a glass test surface. We are setting up a similar experiment with EVA suit fabric.

Bibliography Type:	Description: (Last Updated: 01/23/2025)
Articles in Peer-reviewed Journals	V. Nosenko, J. Goree, Z.W. Ma, D.H.E. Dubin and A. Piel "Compressional and Shear Wakes in a 2D Dusty Plasma Crystal" Physical Review E Vol. 68, 056409-1 – 056409-15, 2003 , Nov-2003
Articles in Peer-reviewed Journals	B. Liu, K. Avinash and J. Goree "Transverse Optical Mode in a One-Dimensional Chain" Physical Review Letters, Vol. 91, 255003, 2003 , Jan-2003
Articles in Peer-reviewed Journals	V. Nosenko, K. Avinash, J. Goree and B. Liu "Nonlinear Interaction of Compressional Waves in a 2D Dusty Plasma Crystal" Physical Review Letters, Vol. 92, 085001-1– 085001-4, 2004 , Feb-2004
Articles in Peer-reviewed Journals	V. Nosenko, K. Avinash, J. Goree and B. Liu "Nonlinear Interaction of Compressional Waves in a 2D Dusty Plasma Crystal" Physical Review Letters, Vol. 92, 085001-1– 085001-4, 2004 , Feb-2004
Articles in Peer-reviewed Journals	B. Liu, K. Avinash and J. Goree "Characterization of Interparticle Potential from Lattice Structure" Physical Review E, Vol. 69, 036410, 2004 , Mar-2004
Articles in Peer-reviewed Journals	K. Avinash, P. Zhu, V. Nosenko and J. Goree "Nonlinear Compressional Waves in a Two-Dimensional Yukawa Lattice" Physical Review E, Vol. 68, 046402-1 - 046402-8 2003 , Oct-2003
Articles in Peer-reviewed Journals	Robert L. Merlino and John A. Goree "Dusty Plasmas in Industry, the Laboratory and Space" Physics Today, Vol. 57, 32-38, 2004 , Jul-2004
Articles in Peer-reviewed Journals	V. Nosenko and J. Goree "Shear Flows and Shear Viscosity in a Two-Dimensional Yukawa System (Dusty Plasma)" Physical Review Letters, Vol. 93, pp. 155004-1 - 155004-4, 2004 , Oct-2004
Presentation	J. Goree and V. Nosenko "Electrostatic Release of Fine Particles Adhered to Surfaces on the Moon or Mars" Conference-Workshop on Strategic Research to Enable NASA's Conference-Workshop on Strategic Research to Enable NASA's Exploration Missions Cleveland, OH Jun-2004
Presentation	V. Nosenko and J. Goree "Shear viscosity measurements in a liquid strongly-coupled dusty plasma" 31st European Physical Society Conference on Plasma Physics, London, UK Jun-2004
Presentation	J. Goree "Voids imaged under microgravity conditions" 31st European Physical Society Conference on Plasma Physics, London, UK Jun-2004
Presentation	V. Nosenko, K. Avinash, J. Goree, and B. Liu "Nonlinear mixing of compressional waves in a 2D dusty plasma crystal" 31st European Physical Society Conference on Plasma Physics, London, UK Jun-2004

Presentation	B. Liu and J. Goree "Modes in a one-dimensional dusty plasma chain" 31st European Physical Society Conference on Plasma Physics, London, UK Jun-2004
Presentation	Jerome Fung, Bin Liu, John Goree, Vladimir Nosenko "Measuring particle charge in an rf dusty plasma" 46th Annual Meeting of the APS Division of Plasma Physics, Savannah, GA Nov-2004
Presentation	Bin Liu and John Goree "MD simulation of a 2-D liquid dusty plasma" 46th Annual Meeting of the APS Division of Plasma Physics, Savannah, GA Nov-2004
Presentation	John Goree "Making a plasma act like a crystal" Physics Dept. Colloquium, Swarthmore College Sep-2003
Presentation	John Goree "Making a plasma act like a crystal" Physics Dept. Colloquium, California State University, Sacramento Oct-2003
Presentation	V. Nosenko, K. Avinash, J. Goree and B. Liu "Nonlinear mixing of compressional waves in a 2D dusty plasma crystal" 56 th Gaseous Electronics Conference, San Francisco, CA Oct-2003
Presentation	J. Goree, B. Liu and K. Avinash "Transverse optical mode in a one-dimensional dusty plasma lattice" 56 th Gaseous Electronics Conference, San Francisco, CA Oct-2003
Presentation	B. Liu, J. Goree, K. Avinash "Sheath diagnostics using particles in a dusty plasma" 56 th Gaseous Electronics Conference, San Francisco, CA Oct-2003
Presentation	Bin Liu, John Goree, Khare Avinash "Transverse optical mode in a one-dimensional dusty plasma lattice" 45th Annual Meeting of the APS Division of Plasma Physics, Albuquerque, NM Oct-2003